AttentivU

Hackathon-oriented paper summary

Kosmyna & Maes, Sensors 2019, 19, 5200

Core contribution.	AttentivU is a model-free closed-loop EEG system. It estimates sustained engagement from a hand-designed spectral ratio, calibrates that score to the individual, and delivers a one-second vibrotactile cue when engagement remains low. Across two learning studies with 48 adults, true EEG-triggered feedback was associated with higher EEG-based engagement and better immediate comprehension than random or absent feedback.

System architecture

The system had two components: a BrainCo Focus 1 EEG headband for real-time state estimation and a scarf containing two vibration motors for feedback. The headband sampled EEG at **160 Hz** and transmitted it over Wi-Fi to a Mac application.

Electrodes are not channels

An **electrode** is a physical conductive contact. A hydrogel electrode uses a water-rich conductive polymer to improve scalp contact. A **channel** is the voltage-difference signal produced by an amplifier:

$$x_{Fpz}(t) = V_{Fpz}(t) - V_{ref}(t)$$

The 10-20 system specifies standardized **scalp locations** such as Fpz and TP9. A location becomes an electrode site when an electrode is placed there; the channel definition additionally depends on the reference or montage.

The BrainCo device contained **three hydrogel electrodes but one EEG channel**: one recording electrode at Fpz, plus reference and ground contacts reported by the paper at TP9. Ground supports stable amplifier operation and does not form an additional EEG channel.

Hardware distinction.	The TP9-TP10 pair mentioned later in the discussion belongs to the separate Smith Lowdown Focus glasses proposed as future hardware, not to the BrainCo Focus 1 used in the experiments.

Engagement and vigilance index

The method assumes that beta power increases relative to alpha and theta during attentive mental activity, whereas increases in alpha and theta are associated with lower vigilance or alertness. Its exact bands were:

Band	Range	Interpretation used by the paper
Theta	4-7 Hz	Higher power contributes to a lower index; associated with reduced vigilance.
Alpha	7-11 Hz	Higher power contributes to a lower index; synchronization may accompany rest.
Beta	11-20 Hz	Higher power contributes to a higher index; associated with attentive cortical activity.

From the power spectral density (PSD), averaged within each band over a five-second sliding window, the system computed

$$E = P_\beta / (P_\alpha + P_\theta)$$

The careful interpretation is therefore: **higher beta power relative to alpha plus theta produces a higher estimated engagement/vigilance score**. The paper did not normalize each band by total spectral power before taking this ratio.

After exponential smoothing, the score was calibrated separately for each participant:

$$E_{norm} = (E_{smooth} - E_{min}) / (E_{max} - E_{min}) \times 100$$

E_{min} represented low engagement, such as relaxation, and E_{max} high engagement, such as solving arithmetic problems. Scores 0-30 were classified as low, 31-70 as medium, and 71-100 as high.

Real-time processing pipeline

#	Operation	Purpose or important detail
1	Raw EEG at 160 Hz	Single Fpz-referenced time series received over Wi-Fi.
2	60 Hz notch, 10 Hz bandwidth	Suppress 60 Hz power-line contamination.
3	4-20 Hz band-pass	Retain the theta-to-beta range and suppress frequencies outside it.
4	Downsample	Reduce temporal data rate; the final sampling rate was not reported.
5	Repeat 4-20 Hz band-pass	Additional post-downsampling filtering, described as noise avoidance.
6	Mean-center and divide by 8	Center and rescale the signal before clipping.
7	Clip to [−4, 4]	Limit large K-complex-like spikes attributed to minor motion artefacts.
8	PSD, 5 s sliding windows	Estimate frequency power from the recent signal. Window overlap and hop were not reported.
9	Average band powers	Obtain $P\theta$, $P\alpha$, and $P\beta$.
10	Compute E	Form the beta/(alpha + theta) ratio.
11	Exponentially weighted moving average	Track sustained trends and suppress transient/movement effects.
12	Individual calibration	Map Esmooth to a participant-specific 0-100 scale.
13	Threshold and feedback	Vibrate when the score remains low for at least 15 s.

All reported filters were **IIR Butterworth filters**. The authors state that the processing pipeline followed Hassib et al.'s EngageMeter (2017) and Szafir and Mutlu's Pay Attention! (2012). The engagement index originated with Pope, Bogart, and Bartolome (1995).

Why divide by 8?

The paper gives no scientific or device-specific justification for the number 8. It appears to be a numerical scaling choice that makes the subsequent [−4, 4] clipping range meaningful or matches the inherited implementation. It should not be copied to ANT data without checking its units and amplitude scale.

Without clipping, dividing the time signal by 8 divides every PSD band power by $8^2 = 64$, which cancels in the ratio:

$$(P_\beta/64) / (P_\alpha/64 + P_\theta/64) = P_\beta / (P_\alpha + P_\theta)$$

Thus scaling alone does not change E; it matters through numerical behavior and which samples are clipped.

Why downsample one channel?

Channel count is spatial; sampling rate is temporal. One channel at 160 Hz still generates 160 values per second. Because the highest retained frequency was 20 Hz, anti-alias filtering followed by a sampling rate above 40 Hz could preserve the frequency range of interest while reducing computation. Nevertheless, one channel at 160 Hz is computationally trivial, so downsampling was not strictly necessary. It was likely retained for efficiency or consistency with earlier pipelines. The pre-downsampling low-pass component is essential; downsampling itself does not remove noise.

Timing and latency

Band power was computed from **five-second sliding windows**. The ratio was then exponentially smoothed over an empirically selected **approximately 15-second decision timescale**, and feedback required engagement to remain low for at least 15 seconds. In the pilot, distraction-related drops appeared with about a 15-second onset.

The exact end-to-end latency cannot be reconstructed because the paper omits the window hop/overlap, EWMA coefficient, output-update frequency, final downsampled rate, and exact relationship between smoothing and the 15-second persistence rule.

> **Interpretation for NOVA.** AttentivU detects sustained engagement changes. Its five-second spectral window and approximately 15-second response scale are too slow for an instantaneous or sub-second pre-action prediction without redesign.

Experimental evidence

Two studies involved **48 adults** in video or face-to-face learning. The conditions were:

- **Biofeedback:** vibration triggered by a detected medium-to-low engagement transition.
- **Random feedback:** vibration unrelated to EEG state.
- **No feedback:** EEG was recorded but no vibration was delivered.

The EEG-triggered condition produced higher EEG-based engagement scores and better immediate comprehension-test performance than random or no feedback. These findings concern **momentary** learning performance; the study did not establish long-term learning gains, lasting improvement in attention, or causal validity of the EEG ratio as a complete measure of engagement.

Four future challenges

Challenge	Paper's concern	NOVA relevance
1. Form factor	The headband was conspicuous and could be lighter; alternatives included around-ear EEG and multimodal glasses.	Out of scope: use the provided ANT hardware rather than redesigning the headset.
2. Motion artefacts	Consumer EEG is movement-sensitive; the experiments deliberately restricted movement.	Constraint to mitigate: use a seated, low-motion task and monitor signal quality. A full hardware solution is infeasible.
3. Setup	The controlled lectures lacked note-taking, questions, normal motion, long-term testing, and broad generalization.	Partly feasible: design a realistic precision task with controlled movement and clear trial-level outcomes.
4. Detection	A fixed ratio may omit relevant state information; future work could test other features, classifiers, fatigue/mood, and EOG.	Most actionable: compare the ratio with additional EEG features or a lightweight model and calibrate within participant.

Direct lessons for our buildathon

1. **Useful architecture:** AttentivU provides a clear precedent for signal acquisition → spectral features → calibrated state → feedback.
2. **No ML is inherently required:** the original system used a formula, individual calibration, and thresholds rather than a trained model.
3. **Performance prediction is still missing:** the NOVA challenge requires correlating neural signals with real-time task performance. We must link pre-action features to trial-level success/failure, or demonstrate that the estimated vigilance state predicts performance in our chosen task.
4. **Timing must be redesigned:** event-aligned pre-action epochs or substantially shorter windows are needed if feedback concerns an upcoming precision action rather than sustained attention.
5. **Do not transplant arbitrary constants:** the factor of 8, clipping range, thresholds, and downsampling rate must be re-established for ANT Neuro data and our task.
6. **Prevent circular validation:** a higher E after feedback shows a change in the same metric that triggered feedback. Behavioral success provides the necessary external outcome.

References

Kosmyna, N., & Maes, P. (2019). AttentivU: An EEG-based closed-loop biofeedback system for real-time monitoring and improvement of engagement for personalized learning. Sensors, 19(23), 5200.

Hassib, M., Schneegass, S., Eiglsperger, P., Henze, N., Schmidt, A., & Alt, F. (2017). EngageMeter: A system for implicit audience engagement sensing using electroencephalography. CHI 2017, 5114-5119.

Szafir, D., & Mutlu, B. (2012). Pay attention! Designing adaptive agents that monitor and improve user engagement. CHI 2012, 11-20.

Pope, A. T., Bogart, E. H., & Bartolome, D. S. (1995). Biocybernetic system evaluates indices of operator engagement in automated task. Biological Psychology, 40, 187-195.